

CineCompose: A Cinematographer’s Assistant to Compose a Framing Correctly (8-Way Move + Signed Rotation Regression)

Multi-Task Transfer Learning for 8-Way Framing Correction and Signed Rotation Regression

1st Al-Mahmud
Department of CSE
BRAC University
Dhaka, Bangladesh
al.mahmud@g.bracu.ac.bd

2nd Reduan Nur Labid
Department of CSE
BRAC University
Dhaka, Bangladesh
reduan.nur.labid@g.bracu.ac.bd

3rd Md Mamun Molla
Department of CSE
BRAC University
Dhaka, Bangladesh
md.mamun.molla@g.bracu.ac.bd

Abstract—CineCompose is a framing assistant designed to provide actionable camera-composition guidance by producing two kinds of corrections: an 8-way movement suggestion (left/right/up/down plus diagonals, with “no move”) and a signed rotation correction (clockwise/counterclockwise degrees) to fix tilt or horizon errors. Instead of relying on expensive manual labeling, we create supervision through a perturbation-based dataset pipeline. Each professional still is treated as a correct “good” reference frame, and multiple “bad” variants are synthesized using controlled directional crops and signed rotations. A multi-head transfer learning model is trained to simultaneously predict movement correction, rotation correction, and a good-vs-bad decision that enables “no suggestion” behavior when an image is already well composed. We conduct experiments using multiple pretrained CNN backbones and compare them using a held-out test split. Our final test evaluation shows EfficientNetB0 achieves the best overall performance with test loss 1.793, move accuracy 0.456 (crop-only), rotation MAE 4.20° (rotate-only), and good/bad accuracy 0.851. Training/validation behavior indicates overfitting typically begins after early epochs, largely due to synthetic perturbation bias and ambiguity in defining a single “correct” correction in some scenes. We conclude with recommendations for making the assistant more composition-aware and robust for real-world deployment.

Index Terms—Cinematography, Composition Assistance, Camera Framing, Multi-Task Learning, Transfer Learning, Rotation Regression, Data Augmentation

I. INTRODUCTION

Strong composition and framing are fundamental to cinematography, yet maintaining them reliably during real-time shooting is difficult, especially for beginners. Even minor errors such as excessive headroom, subject drift, or a slightly tilted horizon can reduce perceived production quality. Traditional composition aids like grids and rule-of-thirds overlays are helpful but passive: they do not explicitly tell a camera operator what adjustment to make.

CineCompose aims to provide direct and interpretable correction commands similar to those used on a real film set. Instead of simply scoring an image as “good” or “bad,” CineCompose outputs actionable suggestions: where to move

the camera (in eight directions) and how much to rotate it (signed degrees), while also allowing the system to output “no suggestion” when correction is unnecessary.

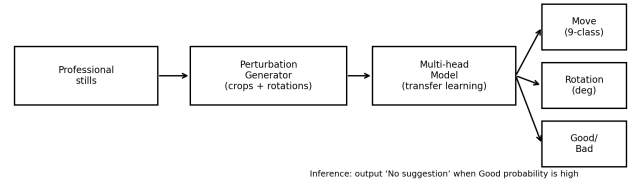


Fig. 1. System overview pipeline.

II. LITERATURE REVIEW

This literature review groups prior work into four relevant themes.

A. Learning from Human Cinematographers

Prior learning-based research demonstrates that professional footage can encode framing behavior. Huang et al. [1] learn camera motion paths from expert recordings, showing the feasibility of predicting camera motion consistent with subject framing and scene context. Jiang et al. [2] propose example-driven camera behavior transfer, reinforcing that camera patterns can be learned from film references. However, these systems typically produce continuous outputs rather than discrete, interpretable commands that motivates CineCompose’s 8-way action space.

B. Reinforcement Learning for Camera Framing

Reinforcement learning (RL) approaches show strong potential for automated camera control. Passalis et al. [3] demonstrate RL-driven drone cinematography using discrete action choices that resemble step-wise camera motion. Wang et al. [4] incorporate saliency-based rotation feedback for 360° videos, which is relevant to continuous rotation correction. Other

RL work [5] emphasizes that reward shaping and semantic understanding can drive camera behavior and aesthetics.

C. Framing-Aware Multimodal Modeling

Multimodal camera models emphasize that framing decisions benefit from both environmental context and subject motion. Courant et al. [8] propose framing-aware multimodal modeling of camera and human motion. Diffusion-based work by Castro et al. [9] suggests camera angle generation from semantic cues, indicating that intent can guide pose adjustments. A survey on trajectory generation [10] highlights a trend toward hybrid frameworks combining discrete actions, continuous motion, and semantic reasoning that supports the design direction of CineCompose.

D. Pose Regression and Rotation Prediction

Pose regression research supports reliable prediction of continuous angles. Naseer et al. [12] demonstrate deep regression for camera pose estimation, while Wang et al. [11] focus on robust regression-based rotation prediction, reinforcing the feasibility of signed rotation regression for framing correction.

Identified Gap: Prior research in computational cinematography includes learning camera patterns from professional film footage, reinforcement learning for camera control in drones and virtual environments, and regression-based models that estimate camera orientation or pose. Many aesthetic models exist as well, but they typically output a quality score rather than corrective instructions. CineCompose is positioned as a practical assistant: it provides an explicit “what to do next” output, discrete movement correction, plus signed rotation regression, making it suitable for real-time guidance. However, this also introduces a key challenge: correction labels must be available and consistent, which motivates our perturbation-based approach for generating supervision.

III. METHODOLOGY

A. Perturbed Dataset Construction

A major challenge in training a framing assistant is obtaining ground-truth “correction labels” from real cinematography. Manual labeling is expensive and subjective because multiple framings may be acceptable depending on narrative intent. CineCompose avoids large-scale manual annotation by creating a perturbed dataset from professional stills. For each `shot_id` folder, we treat the original professional image (`original.jpg`) as the correct reference and label it as GOOD, with the movement target set to “No move” and rotation target set to 0° . We then generate multiple BAD versions using two families of controlled transformations.

Directional crops are applied to simulate framing translation errors. These crops shift the effective framing in one of eight directions (left/right/up/down and diagonals). Each resulting image is labeled with the inverse movement required to correct the framing, so the model learns to map a degraded framing back toward the professional reference. Signed rotations are applied to simulate horizon tilt and camera roll errors, and each rotated image is labeled with the signed degree correction

needed to restore the original orientation. Labels and transformation metadata are stored in a JSON file (`labels.json`) in each shot folder, in a format similar to professional ML datasets.



Fig. 2. Perturbation set per shot: original with 8 crops and +/- rotation

B. Rotation Without Black Borders (Minimal Necessary Scaling)

Naively rotating an image often introduces black corners. If black corners appear only in rotation-perturbed samples, the model may learn a shortcut cue (“black corners means rotation error”) rather than learning the true visual cues related to tilt and horizon. To reduce this risk, our rotation pipeline applies minimal scaling after rotation so the output image remains fully filled, removing border artifacts while keeping scaling as small as possible. This makes the rotation perturbation more realistic and improves the fairness of learning.



Fig. 3. Rotation perturbation: remove black borders via minimal necessary scaling

C. Shot-Level Split (Folder as the Unit)

To ensure a realistic evaluation, we split the dataset at the shot level: each `shot_id` folder is assigned entirely to training, validation, or test. This prevents near-duplicate leakage, where the model might see a perturbed variant of a shot during training and then be evaluated on another variant of the same shot during validation/testing. Shot-level splitting is critical for compositional learning tasks because perturbations are highly correlated with their original reference.

IV. MODEL ARCHITECTURE AND TRAINING

A. Multi-Head Transfer Learning Model

CineCompose uses a transfer learning strategy: a pretrained CNN backbone serves as a shared feature extractor and feeds three task-specific heads. The first head predicts a 9-class

movement output (eight correction directions plus “no move”). The second head predicts a signed rotation correction as a continuous regression output in degrees. The third head predicts whether the frame is already good, enabling the system to output “no suggestion” and avoid unnecessary corrections. This multi-head design makes the system practical and interpretable: the output naturally maps to actions a camera operator can take.

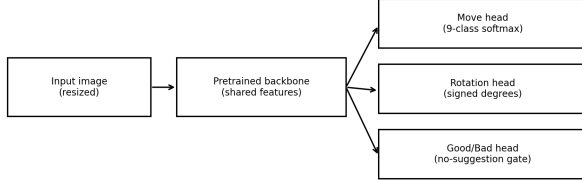


Fig. 4. Multi-Head Model Architecture.

B. Loss Functions and Metrics

Movement prediction is optimized using cross-entropy loss. Rotation prediction uses a robust regression loss (SmoothL1/Huber), which helps reduce sensitivity to outliers. Good/bad prediction uses BCEWithLogits, optionally with a class imbalance weight. Metrics are computed in a task-consistent way: movement accuracy is measured only on crop-perturbed samples, rotation MAE is measured only on rotation-perturbed samples, and good/bad accuracy is measured across all samples. This prevents misleading metrics (e.g., reporting movement accuracy on samples where no move label is relevant).

V. EXPERIMENTS AND EVALUATION

A. Backbones Evaluated

We trained the same multi-head system using several pre-trained CNN backbones and saved the best validation checkpoint for each: ResNet18, ResNet34, MobileNetV3Small, EfficientNetB0, and ConvNeXtTiny. This setup allows us to compare how different feature extractors generalize under the same dataset protocol and objectives.

B. Training/Validation Behavior (Overfitting Evidence from Logs)

Across backbones, training loss decreased strongly each epoch, indicating that models learned the synthetic perturbation supervision effectively. However, validation behavior reveals that the best generalization typically occurs early. For example, EfficientNetB0 validation loss improves from 3.142 $\rightarrow$ 3.094 (epoch 1 $\rightarrow$ 2) and then increases to 3.624 at epoch 3, even though training loss continues decreasing (3.527 $\rightarrow$ 2.643 $\rightarrow$ 2.087). ConvNeXtTiny shows an even clearer effect: validation loss increases sharply by epoch 3 (3.064 $\rightarrow$ 3.144 $\rightarrow$ 4.380) while training loss improves rapidly (3.357 $\rightarrow$ 2.593 $\rightarrow$ 1.890). This pattern indicates that many models begin to overfit after around two epochs, and additional training

increases confidence on training patterns without improving generalization.

Interestingly, some validation metrics such as good/bad accuracy and move accuracy can still rise even when validation loss worsens (e.g., ConvNeXtTiny’s validation good accuracy reaches 0.900 at epoch 3). This suggests the model may become overconfident or learn shortcut cues that help one head (good/bad) while harming the combined multi-task objective. This is a known behavior in synthetic-perturbation learning, where distinguishing “bad” frames becomes easier than predicting the most useful correction.

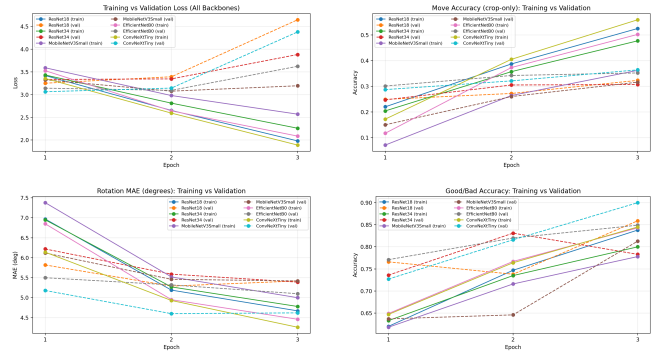


Fig. 5. Training vs Validation Curves. Top Left: Loss. Top Right: Move Acc. Bottom Left: Rot MAE. Bottom Right: Good Acc.

C. Final Test Evaluation (All Best Checkpoints)

After selecting the best checkpoint for each backbone (based on validation selection), we evaluated each model on the held-out test split. The results are summarized in Table I. EfficientNetB0 performs best overall across all reported metrics, including the key “actionable” tasks of movement prediction and rotation correction.

TABLE I
TEST RESULTS SUMMARY (HELD-OUT TEST SPLIT)

Backbone	Test Loss $\downarrow$	Move Acc $\uparrow$	Rot MAE $\downarrow$	Good Acc $\uparrow$
EfficientNetB0	1.793	0.456	4.20°	0.851
ConvNeXtTiny	2.318	0.334	4.72 $^\circ$	0.731
ResNet18	2.373	0.287	5.22 $^\circ$	0.776
MobileNetV3Small	2.447	0.290	4.61 $^\circ$	0.650
ResNet34	2.716	0.253	5.64 $^\circ$	0.746

VI. OVERFITTING ANALYSIS AND REASONING

Overfitting is a central concern in CineCompose because the “bad” examples are generated by a synthetic transformation pipeline rather than collected as real operator mistakes. Even when black borders are removed, crops and rotations can introduce subtle, consistent cues such as resampling patterns, frequency shifts, and systematic changes in texture statistics. Strong networks can learn these cues quickly, especially when trained for multiple epochs. This explains why training loss decreases steadily across all backbones while validation loss often stops improving and then rises by epoch 3. In

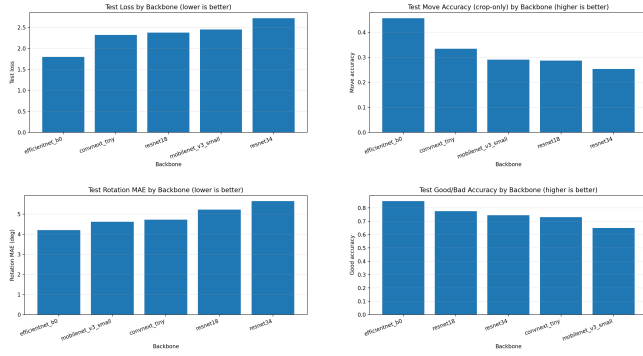


Fig. 6. Test comparison charts across backbones.

other words, the model becomes increasingly specialized in recognizing the training perturbation distribution rather than learning generalizable composition principles.

Additionally, the correction label can be ambiguous in real cinematography: multiple framing choices can be “correct” depending on subject saliency, narrative intent, or style. Our supervision assigns a single corrective direction derived from the inverse transformation, which can introduce label noise. This noise is less harmful for the good/bad head (which mainly needs to separate original from perturbed), but it can degrade the movement and rotation tasks where precision matters. This also helps explain why validation good accuracy may rise even when validation loss worsens: the model becomes confident at recognizing “badness,” but its correction predictions become less calibrated and less consistent overall.

Based on our logs, the most defensible conclusion is that early stopping around epoch 2 is often optimal for this dataset, and training beyond that increases the risk of learning shortcuts. This interpretation is directly supported by EfficientNetB0 and ConvNeXtTiny validation loss behavior, and is broadly consistent across the other models.

VII. CONCLUSION

CineCompose demonstrates a complete pipeline for building an interpretable framing assistant that outputs actionable camera correction guidance. By creating a perturbed dataset from professional stills, we train a multi-head transfer learning model to predict an 8-way movement correction, a signed rotation correction, and a good/bad decision that enables “no suggestion” behavior. Experiments across multiple pretrained CNN backbones show that EfficientNetB0 achieves the best overall generalization on a held-out test split, reaching the lowest test loss and the strongest balance across movement accuracy, rotation MAE, and good/bad accuracy. At the same time, training/validation behavior shows clear signs of overfitting after early epochs, highlighting the importance of early stopping and more realistic supervision for learning true composition principles rather than synthetic artifacts.

VIII. FUTURE WORK

To make CineCompose more robust and composition-aware, future work should focus on improving realism and super-

vision. One direction is to incorporate composition features explicitly (edges, saliency maps, thirds-grid heatmaps) to encourage learning of spatial structure rather than augmentation cues. Another direction is to generate harder and more realistic negatives that resemble real handheld shooting errors (motion blur, compression artifacts, lighting changes), and to add subject-aware perturbations that preserve faces or key objects while shifting composition. Finally, extending the assistant to video requires temporal smoothing so that suggestions remain stable across frames instead of fluctuating moment-to-moment.

REFERENCES

- [1] Y. Huang et al., “Learning to Film from Professional Human Motion Videos,” *CVPR*, 2019.
- [2] Y. Jiang et al., “Example-Driven Virtual Cinematography by Learning Camera Behaviors,” *ACM TOG*, 2020.
- [3] N. Passalis et al., “Deep Reinforcement Learning for Automatic Cinematography with Drones,” *IEEE*, 2019.
- [4] Y. Wang et al., “Attention-Based Deep RL for Virtual Cinematography of 360° Videos,” *IEEE TMM*, 2020.
- [5] Z. Yu et al., “Reinforcement Learning-Based Cinematography for Animation,” *IEEE MIPR*, 2022.
- [6] O. Xie et al., “GAIT: Generating Aesthetic Indoor Tours,” *ICCV*, 2023.
- [7] J. Zhou et al., “Automatic Camera Trajectory Control with Enhanced Immersion,” *IEEE*, 2024.
- [8] J. Courant et al., “Framing-Aware Multimodal Camera and Human Motion Modeling,” 2025.
- [9] M. Castro et al., “A Cinematographic Camera Diffusion Model,” 2024.
- [10] L. Xie et al., “Camera Trajectory Generation: A Comprehensive Survey,” 2025.
- [11] M. Wang et al., “Robust Regression-Based Camera Rotation Prediction,” *IEEE TIP*, 2018.
- [12] T. Naseer et al., “Deep Regression for Camera Pose Estimation,” *CVPR*, 2017.